Riad Mahmud Chowdhury

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CAREER OBJECTIVE

My goal is to expand my research and technical capabilities in Data Science, Machine Learning (ML), and Cyber security across the end-to-end data lifecycle—from data preprocessing and exploratory analysis to model deployment. I aim to apply these skills toward cutting-edge research in cybersecurity while actively investigating methods to make machine learning workflows more energy-efficient and environmentally sustainable.

RESEARCH INTERESTS

Machine Learning (ML), **Data Science**, **Cybersecurity**

EDUCATION

Nantong University, School of Artificial Intelligence and Computer Science — **Nantong, Jiangsu, China**
Bachelor of Science Degree in Computer Science and Technology — 2023.09.10 – Present (Exp. 2027)
CGPA: **4.00/4.00** (Completed 6th Semester / 144 Credits)
Advisor / Supervisor: **Haochen Wang (王皓晨)**

Chattogram Bandar College — **Chattogram, Bangladesh**
Higher Secondary Certificate (HSC) in Science — 2018 – 2020
Result: GPA: **4.87 / 5.00**

Mandarin Language Proficiency (HSK Level 4) — **Nantong University, China**
Result: **220 / 300** — 2025.12

SKILLS SUMMARY

- **Programming:** Python, R, JavaScript, C#, Java, SQL, HTML/CSS
- **Data Science & AI/ML:** Machine Learning, Data Preprocessing, Exploratory Data Analysis (EDA), Data Visualization, Pandas, NumPy, Scikit-learn, Matplotlib, Plotly, Tidyverse, Ggplot2
- **Tools & Platforms:** Visual Studio Code, Google Colab, Jupyter Notebook, Git, GitHub, LaTeX, RStudio, Zotero, Microsoft Office
- **Languages:** Bengali (Native), English (Advanced / Fluent), Mandarin (Intermediate)

CONFERENCE PUBLICATIONS

- **International Conferences:**
  - **[IEEE CSDE'26]:** Md Mostofa Nurannabi Shakil, **Riad Mahmud Chowdhury**, Tanvir Hossain, Md. Alamgir Hossain, Md. Samiul Islam, and Md. Shafiqul Islam. 'Provable Moving-Target Defense for Cyber-Physical Networks: From the Security-Performance Frontier to No-Regret Defense'. In Proceedings of the IEEE International Conference on Computer Science and Data Engineering (IEEE CSDE 2026). (Accepted, 2026)

RESEARCH EXPERIENCE

Research Contributor (Data Analysis) — IEEE CSDE 2026

- Assisted extensively in the experimental analysis and framework evaluation of the research project.
- Contributed to the research manuscript preparation and formatting for the final submission package.
- Collaborated with cross-institutional authors to validate data outcomes and model performance.

PROJECTS & SOFTWARE DEVELOPMENT

• [Banking-Web-C-] Online Banking Web System Dec 2025

Architected a secure full-stack web application using C# implementing robust object-oriented transaction logic, customer account management, and database connectivity. Emphasized backend data integrity, input sanitation, and modular design.

• [lego-data-analysis] Multi-decade Structural Dataset Insights 2026

Developed an end-to-end data analysis pipeline in Jupyter Notebook using Python and Pandas to parse multi-table relational schema. Extracted long-term component growth trends, part complexity, and historical color distribution patterns.

• [StackOverflow-Language-Trends] Developer Adoption Analytics 2026

Created a time-series exploratory pipeline utilizing Python, Pandas, and Matplotlib to analyze multi-year programming language popularity, developer discussion frequencies, and evolving open-source developer tooling trends.

• [college-salaries-analysis-pandas] Post-Graduation Compensation Modeling 2026

Engineered quantitative models using Pandas and NumPy to clean and aggregate higher-education wage demographics. Analyzed career earnings trajectories, collegiate institution tiers, and financial payback periods across diverse majors.

• [Superstore-Sales-Profitability-Analysis] Enterprise Retail Analytics 2026

Designed an exploratory analytics framework using R (Tidyverse & Ggplot2) to cleanse corporate sales records, analyze discount sensitivity, visualize regional logistics overhead, and highlight product line profitability margins.

CERTIFICATIONS & PROFESSIONAL TRAINING

• Data, Data, Everywhere — Google Data Analytics Professional Certificate Coursera | 2026

Credential: Certificate Verification Link [Verified / Completed]

• 100 Days of Code: The Complete Python Pro Bootcamp Udemy | 2026

Credential: Certificate Verification Link [Verified / Completed]

ACADEMIC HONORS & EXTRA-CURRICULARS

- **Perfect Academic Standing (CGPA 4.00 / 4.00):** Ranked in top tier throughout all completed semesters at Nantong University, School of Artificial Intelligence and Computer Science (2023 – Present).
- **Academic Mentorship:** Assisted peer student groups in scientific python environments, structured notebook practices, and data visualization setups.

ACADEMIC REFERENCES

Haochen Wang (王皓晨)
School of Artificial Intelligence and Computer Science
Nantong University, Nantong, Jiangsu, China
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